# SN74LVC2G34 Dual Buffer Gate

## 1 Features

- Available in the Texas Instruments NanoFree™ Package
- Supports 5.5-V $V_{CC}$ Operation
- Inputs Accept Voltages to 5.5 V
- Maximum $t_{pd}$ of 4.1 ns at 3.3 V
- Low Power Consumption, 10-µA Maximum $I_{CC}$
- ±24-mA Output Drive at 3.3 V
- Typical $V_{OLP}$ (Output Ground Bounce) <0.8 V at $V_{CC}$ = 3.3 V, $T_A$ = 25°C
- Typical $V_{OHV}$ (Output $V_{OH}$ Undershoot) >2 V at $V_{CC}$ = 3.3 V, $T_A$ = 25°C
- $I_{off}$ Supports Live Insertion, Partial-Power-Down Mode, and Back-Drive Protection
- Can Be Used as a Down Translator to Translate Inputs From a Maximum of 5.5 V Down to the $V_{CC}$ Level
- Latch-Up Performance Exceeds 100 mA Per JESD 78, Class II
- ESD Protection Exceeds JESD 22
  - 2000-V Human Body Model (A114-A)
  - 200-V Machine Model (A115-A)
  - 1000-V Charged-Device Model (C101)

## 2 Applications

- AV Receivers
- Audio Docks: Portable
- Blu-ray Player and Home Theaters
- DVD Recorders and Players
- Embedded PCs
- MP3 Players and Recorders (Portable Audio)
- Personal Digital Assistant (PDA)
- Power: Telecom/Server AC/DC Supply: Single Controller: Analog and Digital
- Solid-State Drive (SSD): Client and Enterprise
- TV: LCD/Digital and High-Definition (HDTV)
- Tablets: Enterprise
- Video Analytics: Servers
- Wireless Headsets, Keyboards, and Mice

## 3 Description

The SN74LVC2G34 device is a dual buffer gate designed for 1.65-V to 5.5-V $V_{CC}$ operation. The SN74LVC2G34 device performs the Boolean function Y = A in positive logic.

NanoFree package technology is a major breakthrough in IC packaging concepts, using the die as the package.

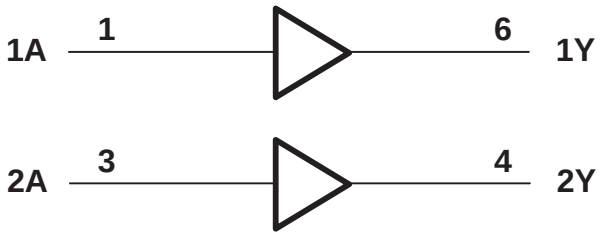
This device is fully specified for partial-power-down applications using $I_{off}$. The $I_{off}$ circuitry disables the outputs, preventing damaging current backflow through the device when it is powered down.

**Device Information(1)**

| PART NUMBER | PACKAGE | BODY SIZE (NOM) |
|---|---|---|
| SN74LVC2G34DBV | SOT-23 (6) | 2.90 mm × 1.60 mm |
| SN74LVC2G34DCK | SC70 (6) | 2.00 mm × 1.25 mm |
| SN74LVC2G34DRL | SOT (6) | 1.60 mm × 1.20 mm |
| SN74LVC2G34YZP | DSBGA (6) | 1.41 mm × 0.91 mm |

(1) For all available packages, see the orderable addendum at the end of the data sheet.

**Simplified Schematic**




An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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## 4 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

**Changes from Revision I (December 2013) to Revision J** **Page**

- Added *Applications*, *Device Information* table, *Pin Configuration and Functions* section, *ESD Ratings* table, *Thermal Information* table, *Typical Characteristics* section, *Feature Description* section, *Device Functional Modes*, *Application and Implementation* section, *Power Supply Recommendations* section, *Layout* section, *Device and Documentation Support* section, and *Mechanical, Packaging, and Orderable Information* section ... 1
- Deleted part number from *Switching Characteristics* table headers. ... 5

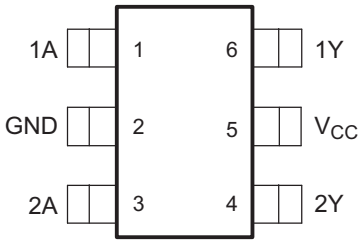
**Changes from Revision H (February 2007) to Revision I** **Page**

- Updated document to new TI data sheet format. ... 1
- Removed *Ordering Information* table. ... 1
- Updated *Features* section ... 1
- Updated operating temperature range. ... 4

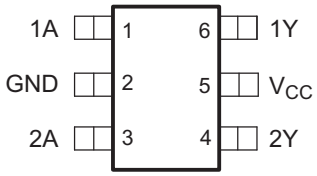

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# 5 Pin Configuration and Functions

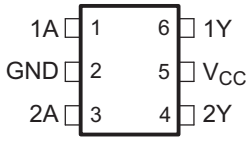
**DBV Package**
**6-Pin SOT-23**
**Top View**




**DCK Package**
**6-Pin SC70**
**Top View**




**DRL Package**
**6-Pin SOT**
**Top View**




**YZP Package**
**6-Pin DSBGA**
**Bottom View**




**Pin Functions**(1)

| PIN | | I/O | DESCRIPTION |
|---|---|---|---|
| NAME | NO. | | |
| 1A | 1 | I | Buffer Input 1 |
| 1Y | 6 | O | Buffer Output 1 |
| 2A | 3 | I | Buffer Input 2 |
| 2Y | 4 | O | Buffer Output 2 |
| GND | 2 | — | Ground pin |
| $V_{CC}$ | 5 | — | Power pin |

(1) See mechanical drawings for dimensions.

# 6 Specifications

## 6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

| | | | MIN | MAX | UNIT |
|---|---|---|---|---|---|
| $V_{CC}$ | Supply voltage | | –0.5 | 6.5 | V |
| $V_I$ | Input voltage(2) | | –0.5 | 6.5 | V |
| $V_O$ | Voltage applied to any output in the high-impedance or power-off state(2) | | –0.5 | 6.5 | V |
| $V_O$ | Voltage applied to any output in the high or low state(2)(3) | | –0.5 | $V_{CC}$ + 0.5 | V |
| $I_{IK}$ | Input clamp current | $V_I$ < 0 | | –50 | mA |
| $I_{OK}$ | Output clamp current | $V_O$ < 0 | | –50 | mA |
| $I_O$ | Continuous output current | | | ±50 | mA |
| | Continuous current through $V_{CC}$ or GND | | | ±100 | mA |
| $T_J$ | Junction temperature | | | 150 | °C |
| $T_{stg}$ | Storage temperature | | –65 | 150 | °C |

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) The input negative-voltage and output voltage ratings may be exceeded if the input and output current ratings are observed.
(3) The value of $V_{CC}$ is provided in the *Recommended Operating Conditions* table.

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## 6.2 ESD Ratings

| | | | VALUE | UNIT |
|---|---|---|---|---|
| $V_{(ESD)}$ | Electrostatic discharge | Human body model (HBM), per ANSI/ESDA/JEDEC JS-001, all pins(1) | 2500 | V |
| | | Charged-device model (CDM), per JEDEC specification JESD22-C101, all pins(2) | 1500 | |

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

## 6.3 Recommended Operating Conditions(1)

| | | | MIN | MAX | UNIT |
|---|---|---|---|---|---|
| $V_{CC}$ | Supply voltage | Operating | 1.65 | 5.5 | V |
| | | Data retention only | 1.5 | | |
| $V_{IH}$ | High-level input voltage | $V_{CC}$ = 1.65 V to 1.95 V | 0.65 × $V_{CC}$ | | V |
| | | $V_{CC}$ = 2.3 V to 2.7 V | 1.7 | | |
| | | $V_{CC}$ = 3 V to 3.6 V | 2 | | |
| | | $V_{CC}$ = 4.5 V to 5.5 V | 0.7 × $V_{CC}$ | | |
| $V_{IL}$ | Low-level input voltage | $V_{CC}$ = 1.65 V to 1.95 V | | 0.35 × $V_{CC}$ | V |
| | | $V_{CC}$ = 2.3 V to 2.7 V | | 0.7 | |
| | | $V_{CC}$ = 3 V to 3.6 V | | 0.8 | |
| | | $V_{CC}$ = 4.5 V to 5.5 V | | 0.3 × $V_{CC}$ | |
| $V_I$ | Input voltage | | 0 | 5.5 | V |
| $V_O$ | Output voltage | | 0 | $V_{CC}$ | V |
| $I_{OH}$ | High-level output current | $V_{CC}$ = 1.65 V | | –4 | mA |
| | | $V_{CC}$ = 2.3 V | | –8 | |
| | | $V_{CC}$ = 3 V | | –16 | |
| | | | | –24 | |
| | | $V_{CC}$ = 4.5 V | | –32 | |
| $I_{OL}$ | Low-level output current | $V_{CC}$ = 1.65 V | | 4 | mA |
| | | $V_{CC}$ = 2.3 V | | 8 | |
| | | $V_{CC}$ = 3 V | | 16 | |
| | | | | 24 | |
| | | $V_{CC}$ = 4.5 V | | 32 | |
| Δt/Δv | Input transition rise or fall rate | $V_{CC}$ = 1.8 V ± 0.15 V, 2.5 V ± 0.2 V | | 20 | ns/V |
| | | $V_{CC}$ = 3.3 V ± 0.3 V | | 10 | |
| | | $V_{CC}$ = 5 V ± 0.5 V | | 5 | |
| $T_A$ | Operating free-air temperature | DBV, DCK, DRL Package | –40 | 125 | °C |
| | | YZP Package | -40 | 85 | |

(1) All unused inputs of the device must be held at $V_{CC}$ or GND to ensure proper device operation. Refer to the TI application report, *Semiconductor and Implications of Slow or Floating CMOS Inputs*, SCBA004.

## 6.4 Thermal Information

| THERMAL METRIC(1) | | SN74LVC2G34 | | | | UNIT |
|---|---|---|---|---|---|---|
| | | DBV (SOT-23) | DCK (SC70) | DRL (SOT) | YZP (DSBGA) | |
| | | 6 PINS | 6 PINS | 6 PINS | 6 PINS | |
| $R_{\theta JA}$ | Junction-to-ambient thermal resistance | 165 | 259 | 142 | 123 | °C/W |

(1) For more information about traditional and new thermal metrics, see the *IC Package Thermal Metrics* application report, SPRA953.


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## 6.5 Electrical Characteristics

over recommended operating free-air temperature range (unless otherwise noted)

| PARAMETER | | TEST CONDITIONS | $V_{CC}$ | MIN | TYP(1) | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| $V_{OH}$ | | $I_{OH}$ = –100 µA | 1.65 V to 5.5 V | $V_{CC}$ – 0.1 | | | V |
| | | $I_{OH}$ = –4 mA | 1.65 V | 1.2 | | | |
| | | $I_{OH}$ = –8 mA | 2.3 V | 1.9 | | | |
| | | $I_{OH}$ = –16 mA | 3 V | 2.4 | | | |
| | | $I_{OH}$ = –24 mA | | 2.3 | | | |
| | | $I_{OH}$ = –32 mA | 4.5 V | 3.8 | | | |
| $V_{OL}$ | | $I_{OL}$ = 100 µA | 1.65 V to 5.5 V | | | 0.1 | V |
| | | $I_{OL}$ = 4 mA | 1.65 V | | | 0.45 | |
| | | $I_{OL}$ = 8 mA | 2.3 V | | | 0.3 | |
| | | $I_{OL}$ = 16 mA | 3 V | | | 0.4 | |
| | | $I_{OL}$ = 24 mA | | | | 0.55 | |
| | | $I_{OL}$ = 32 mA | 4.5 V | | | 0.55 | |
| $I_I$ | A inputs | $V_I$ = 5.5 V or GND | 0 to 5.5 V | | | ±5 | µA |
| $I_{off}$ | | $V_I$ or $V_O$ = 5.5 V | 0 | | | ±10 | µA |
| $I_{CC}$ | | $V_I$ = 5.5 V or GND, $I_O$ = 0 | 1.65 V to 5.5 V | | | 10 | µA |
| $\Delta I_{CC}$ | | One input at $V_{CC}$ – 0.6 V, Other inputs at $V_{CC}$ or GND | 3 V to 5.5 V | | | 500 | µA |
| $C_i$ | | $V_I$ = $V_{CC}$ or GND | 3.3 V | | 3.5 | | pF |

(1) All typical values are at $V_{CC}$ = 3.3 V, $T_A$ = 25°C.

## 6.6 Switching Characteristics

over recommended operating free-air temperature range (unless otherwise noted) (see Figure 2)

| PARAMETER | FROM (INPUT) | TO (OUTPUT) | OPERATING FREE-AIR TEMPERATURE ($T_A$) | $V_{CC}$ | MIN | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| $t_{pd}$ | A | Y | –40°C to 85°C | $V_{CC}$ = 1.8 V ± 0.15 V | 3.2 | 8.6 | ns |
| | | | | $V_{CC}$ = 2.5 V ± 0.2 V | 1.5 | 4.4 | |
| | | | | $V_{CC}$ = 3.3 V ± 0.3 V | 1.4 | 4.1 | |
| | | | | $V_{CC}$ = 5 V ± 0.5 V | 1 | 3.2 | |
| $t_{pd}$ | A | Y | –40°C to 125°C | $V_{CC}$ = 1.8 V ± 0.15 V | 3.2 | 9.6 | ns |
| | | | | $V_{CC}$ = 2.5 V ± 0.2 V | 1.5 | 4.9 | |
| | | | | $V_{CC}$ = 3.3 V ± 0.3 V | 1.2 | 4.6 | |
| | | | | $V_{CC}$ = 5 V ± 0.5 V | 1 | 3.7 | |

## 6.7 Operating Characteristics

$T_A$ = 25°C

| PARAMETER | | TEST CONDITIONS | $V_{CC}$ | TYP | UNIT |
|---|---|---|---|---|---|
| $C_{pd}$ | Power dissipation capacitance | f = 10 MHz | $V_{CC}$ = 1.8 V | 14 | pF |
| | | | $V_{CC}$ = 2.5 V | 14 | |
| | | | $V_{CC}$ = 3.3 V | 15 | |
| | | | $V_{CC}$ = 5 V | 17 | |

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## 6.8 Typical Characteristics

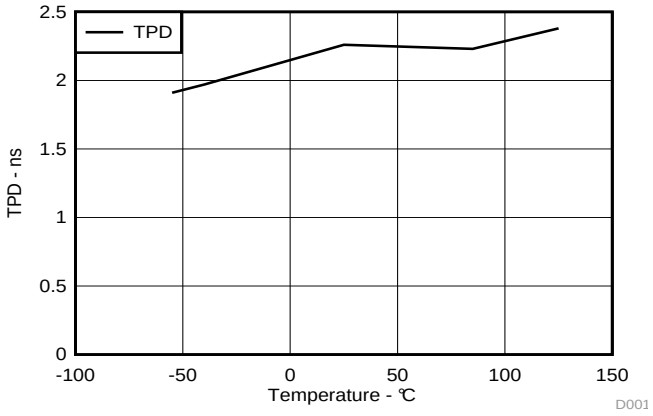
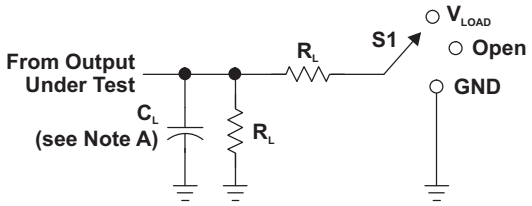



**Figure 1. TPD Across Temperature at 3.3-V $V_{CC}$**


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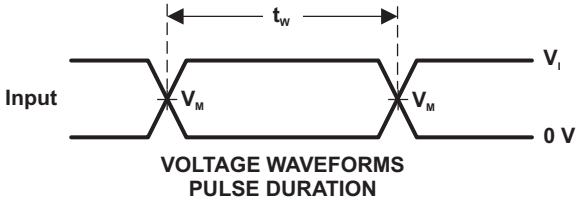
# 7 Parameter Measurement Information



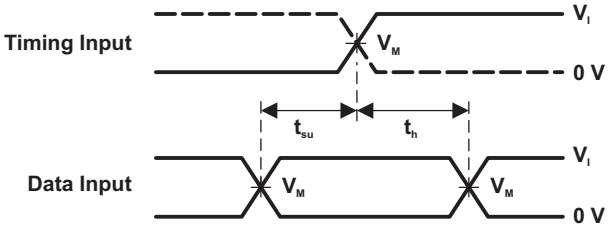

LOAD CIRCUIT

| TEST | S1 |
|---|---|
| $t_{PLH}/t_{PHL}$ | Open |
| $t_{PLZ}/t_{PZL}$ | $V_{LOAD}$ |
| $t_{PHZ}/t_{PZH}$ | GND |

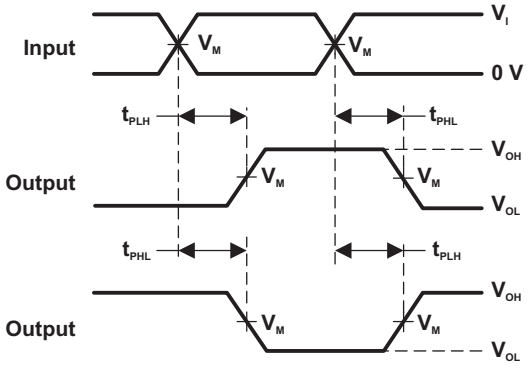
| $V_{CC}$ | INPUTS | | $V_M$ | $V_{LOAD}$ | $C_L$ | $R_L$ | $V_\Delta$ |
|---|---|---|---|---|---|---|---|
| | $V_I$ | $t_r/t_f$ | | | | | |
| 1.8 V ± 0.15 V | $V_{CC}$ | ≤2 ns | $V_{CC}/2$ | 2 × $V_{CC}$ | 30 pF | 1 kΩ | 0.15 V |
| 2.5 V ± 0.2 V | $V_{CC}$ | ≤2 ns | $V_{CC}/2$ | 2 × $V_{CC}$ | 30 pF | 500 Ω | 0.15 V |
| 3.3 V ± 0.3 V | 3 V | ≤2.5 ns | 1.5 V | 6 V | 50 pF | 500 Ω | 0.3 V |
| 5 V ± 0.5 V | $V_{CC}$ | ≤2.5 ns | $V_{CC}/2$ | 2 × $V_{CC}$ | 50 pF | 500 Ω | 0.3 V |



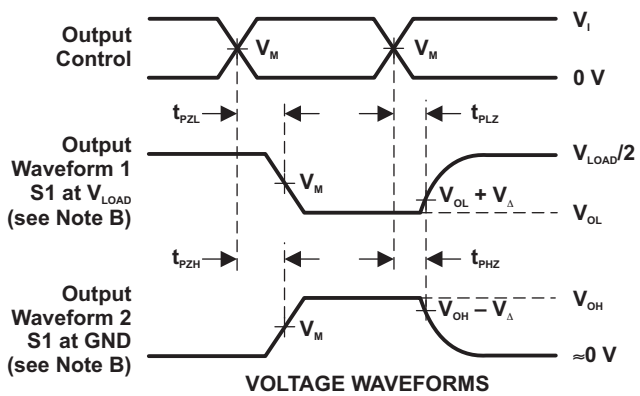

VOLTAGE WAVEFORMS
PULSE DURATION




VOLTAGE WAVEFORMS
SETUP AND HOLD TIMES




VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES
INVERTING AND NONINVERTING OUTPUTS




VOLTAGE WAVEFORMS
ENABLE AND DISABLE TIMES
LOW- AND HIGH-LEVEL ENABLING

NOTES: A. $C_L$ includes probe and jig capacitance.
B. Waveform 1 is for an output with internal conditions such that the output is low, except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high, except when disabled by the output control.
C. All input pulses are supplied by generators having the following characteristics: PRR ≤ 10 MHz, $Z_O$ = 50 Ω.
D. The outputs are measured one at a time, with one transition per measurement.
E. $t_{PLZ}$ and $t_{PHZ}$ are the same as $t_{dis}$.
F. $t_{PZL}$ and $t_{PZH}$ are the same as $t_{en}$.
G. $t_{PLH}$ and $t_{PHL}$ are the same as $t_{pd}$.
H. All parameters and waveforms are not applicable to all devices.

**Figure 2. Load Circuit and Voltage Waveforms**

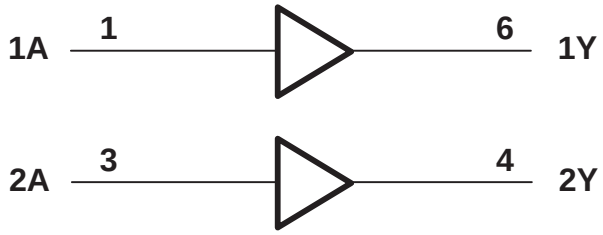
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# 8 Detailed Description

## 8.1 Overview

The SN74LVC2G34 device contains two buffer gates that each perform the Boolean function Y = A. This device is fully specified for partial-power-down applications using $I_{off}$. The $I_{off}$ circuitry disables the outputs, preventing damaging current backflow through the device when it is powered down.

## 8.2 Functional Block Diagram




## 8.3 Feature Description

The SN74LVC2G34 device has a wider operating voltage range, operating from 1.65 V to 5.5 V, and allows down voltage translation. The SN74LVC2G34 $I_{off}$ feature allows voltages on the inputs and outputs when $V_{CC}$ is 0 V.

## 8.4 Device Functional Modes

Table 1 lists the functional modes of the SN74LVC2G34.

**Table 1. Function Table**

| INPUT A | OUTPUT Y |
|---|---|
| H | H |
| L | L |


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# 9 Application and Implementation

## 9.1 Application Information

The SN74LVC2G34 is a high-drive CMOS device that can be used as a buffer with a high output drive, such as an LED application. It can produce 24 mA of drive current at 3.3 V, making it ideal for driving multiple outputs and good for high-speed applications up to 100 MHz. The inputs are 5.5-V tolerant allowing it to translate down to $V_{CC}$.

## 9.2 Typical Application

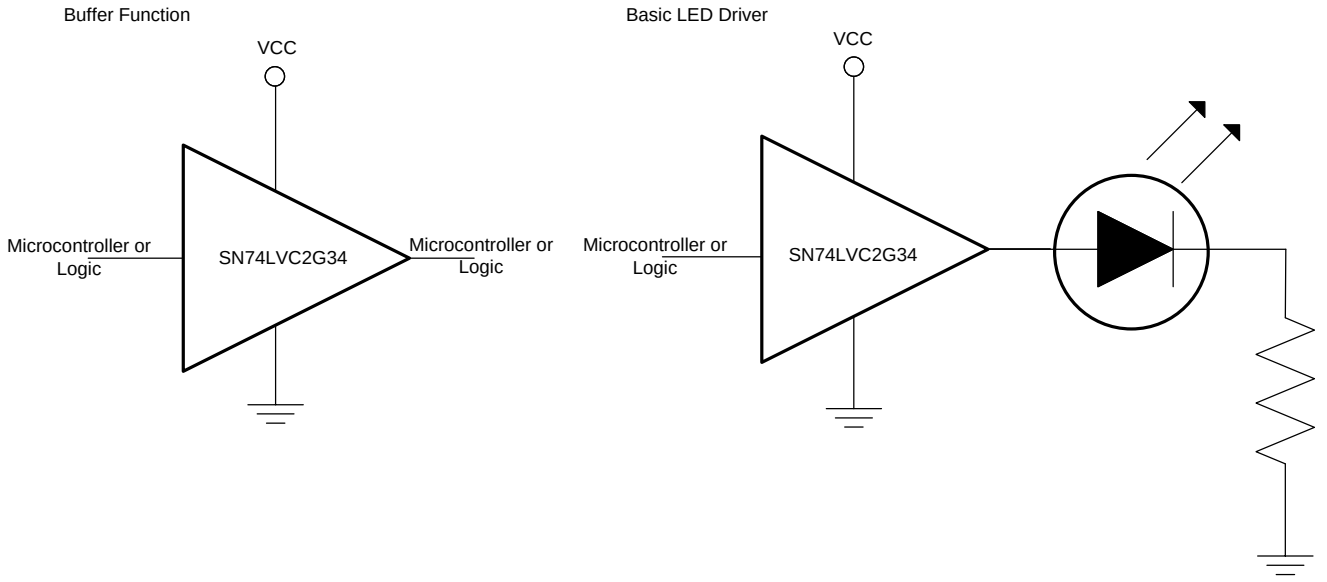



**Figure 3. Typical Application**

### 9.2.1 Design Requirements

This device uses CMOS technology and has balanced output drive. Take care to avoid bus contention because it can drive currents that would exceed maximum limits. The high drive will also create fast edges into light loads so routing and load conditions must be considered to prevent ringing.

### 9.2.2 Detailed Design Procedure

1. Recommended Input Conditions
   - Rise time and fall time specs. See (Δt/ΔV) in the *Recommended Operating Conditions* table.
   - Specified high and low levels. See ($V_{IH}$ and $V_{IL}$) in the *Recommended Operating Conditions* table.
   - Inputs are overvoltage tolerant allowing them to go as high as (VI max) in the *Recommended Operating Conditions* table at any valid $V_{CC}$.
2. Recommended Output Conditions
   - Load currents must not exceed ($I_O$ max) per output and must not exceed (Continuous current through $V_{CC}$ or GND) total current for the part. These limits are located in the *Recommended Operating Conditions* table.
   - Outputs much not be pulled above $V_{CC}$ under normal operating conditions.

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## Typical Application (continued)

### 9.2.3 Application Curve

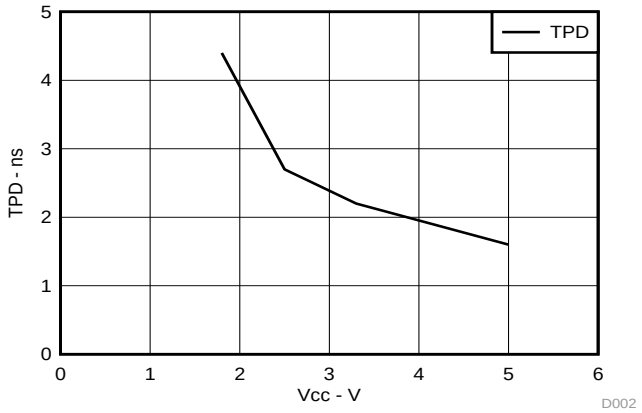



**Figure 4. TPD Across $V_{CC}$ at 25°C**

## 10 Power Supply Recommendations

The power supply can be any voltage between the minimum and maximum supply voltage rating located in the *Recommended Operating Conditions* table.

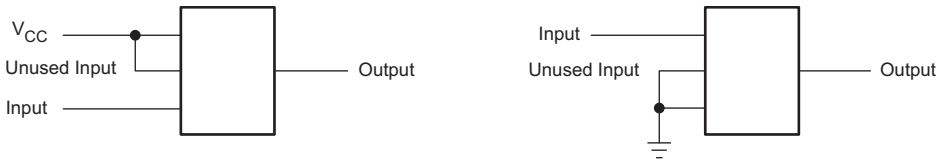
Each $V_{CC}$ pin must have a good bypass capacitor to prevent power disturbance. For devices with a single supply, a 0.1-µF capacitor is recommended and if there are multiple $V_{CC}$ pins then a 0.01-µF or 0.022-µF capacitor is recommended for each power pin. It is ok to parallel multiple bypass caps to reject different frequencies of noise. 0.1-µF and 1-µF capacitors are commonly used in parallel. The bypass capacitor must be installed as close to the power pin as possible for best results.

## 11 Layout

### 11.1 Layout Guidelines

When using multiple bit logic devices inputs must not ever float. In many cases, functions or parts of functions of digital logic devices are unused; for example, when only two inputs of a triple-input AND gate are used or only 3 of the 4 buffer gates are used. Such input pins must not be left unconnected because the undefined voltages at the outside connections result in undefined operational states. Specified below are the rules that must be observed under all circumstances. All unused inputs of digital logic devices must be connected to a high or low bias to prevent them from floating. The logic level that must be applied to any particular unused input depends on the function of the device. Generally they will be tied to GND or $V_{CC}$ whichever make more sense or is more convenient.

### 11.2 Layout Example





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# 12 Device and Documentation Support

## 12.1 Documentation Support

### 12.1.1 Documentation Support

For related documentation, see the following:

*Implications of Slow or Floating CMOS Inputs*, SCBA004

## 12.2 Community Resources

The following links connect to TI community resources. Linked contents are provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's Terms of Use.

**TI E2E™ Online Community** ***TI's Engineer-to-Engineer (E2E) Community.*** Created to foster collaboration among engineers. At e2e.ti.com, you can ask questions, share knowledge, explore ideas and help solve problems with fellow engineers.

**Design Support** ***TI's Design Support*** Quickly find helpful E2E forums along with design support tools and contact information for technical support.

## 12.3 Trademarks

NanoFree, E2E are trademarks of Texas Instruments.
All other trademarks are the property of their respective owners.

## 12.4 Electrostatic Discharge Caution



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

## 12.5 Glossary

SLYZ022 — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

# 13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical packaging and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser based versions of this data sheet, refer to the left hand navigation.

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## PACKAGING INFORMATION

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| SN74LVC2G34DBVR | Active | Production | SOT-23 (DBV) \| 6 | 3000 \| LARGE T&R | Yes | NIPDAU \| SN | Level-1-260C-UNLIM | -40 to 125 | (C345, C34F, C34K, C34O, C34R) |
| SN74LVC2G34DBVR.Z | Active | Production | SOT-23 (DBV) \| 6 | 3000 \| LARGE T&R | Yes | SN | Level-1-260C-UNLIM | -40 to 125 | (C345, C34F, C34K, C34O, C34R) |
| SN74LVC2G34DBVRE4 | Active | Production | SOT-23 (DBV) \| 6 | 3000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | (C34F, C34R) |
| SN74LVC2G34DBVRG4 | Active | Production | SOT-23 (DBV) \| 6 | 3000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | (C34F, C34R) |
| SN74LVC2G34DBVRG4.Z | Active | Production | SOT-23 (DBV) \| 6 | 3000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | (C34F, C34R) |
| SN74LVC2G34DBVT | Active | Production | SOT-23 (DBV) \| 6 | 250 \| SMALL T&R | Yes | NIPDAU \| SN | Level-1-260C-UNLIM | -40 to 125 | (C345, C34F, C34K, C34R) |
| SN74LVC2G34DBVT.Z | Active | Production | SOT-23 (DBV) \| 6 | 250 \| SMALL T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | (C345, C34F, C34K, C34R) |
| SN74LVC2G34DBVTG4 | Active | Production | SOT-23 (DBV) \| 6 | 250 \| SMALL T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | (C34F, C34R) |
| SN74LVC2G34DBVTG4.Z | Active | Production | SOT-23 (DBV) \| 6 | 250 \| SMALL T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | (C34F, C34R) |
| SN74LVC2G34DCK3 | Active | Production | SC70 (DCK) \| 6 | 3000 \| LARGE T&R | Yes | SNBI | Level-1-260C-UNLIM | -40 to 85 | C9Z |
| SN74LVC2G34DCK3.Z | Active | Production | SC70 (DCK) \| 6 | 3000 \| LARGE T&R | Yes | SNBI | Level-1-260C-UNLIM | -40 to 125 | C9Z |
| SN74LVC2G34DCKR | Active | Production | SC70 (DCK) \| 6 | 3000 \| LARGE T&R | Yes | NIPDAU \| SN \| NIPDAU | Level-1-260C-UNLIM | -40 to 125 | (C95, C9F, C9K, C9R) |
| SN74LVC2G34DCKR.Z | Active | Production | SC70 (DCK) \| 6 | 3000 \| LARGE T&R | Yes | SN | Level-1-260C-UNLIM | -40 to 125 | (C95, C9F, C9K, C9R) |
| SN74LVC2G34DCKRE4 | Active | Production | SC70 (DCK) \| 6 | 3000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | C95 |
| SN74LVC2G34DCKRG4 | Active | Production | SC70 (DCK) \| 6 | 3000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | C95 |
| SN74LVC2G34DCKRG4.Z | Active | Production | SC70 (DCK) \| 6 | 3000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | C95 |
| SN74LVC2G34DRLR | Active | Production | SOT-5X3 (DRL) \| 6 | 4000 \| LARGE T&R | Yes | NIPDAU \| NIPDAUAG | Level-1-260C-UNLIM | -40 to 125 | (1K8, C97, C9R) |
| SN74LVC2G34DRLR.Z | Active | Production | SOT-5X3 (DRL) \| 6 | 4000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | (1K8, C97, C9R) |
| SN74LVC2G34DRLRG4.Z | Active | Production | SOT-5X3 (DRL) \| 6 | 4000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 125 | 1K8 |
| SN74LVC2G34YZPR | Active | Production | DSBGA (YZP) \| 6 | 3000 \| LARGE T&R | Yes | SNAGCU | Level-1-260C-UNLIM | -40 to 85 | C9N |
| SN74LVC2G34YZPR.Z | Active | Production | DSBGA (YZP) \| 6 | 3000 \| LARGE T&R | Yes | SNAGCU | Level-1-260C-UNLIM | -40 to 125 | C9N |

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

**Important Information and Disclaimer:**The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

**OTHER QUALIFIED VERSIONS OF SN74LVC2G34 :**

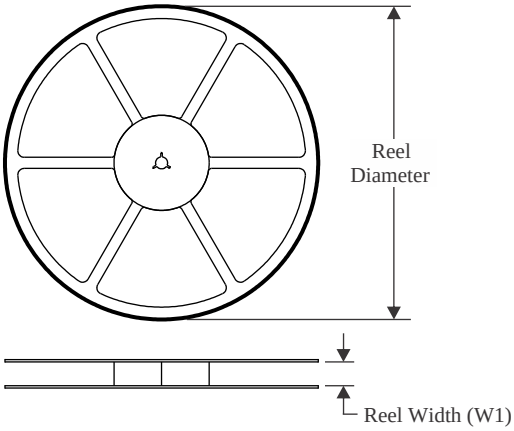
- Enhanced Product : SN74LVC2G34-EP

NOTE: Qualified Version Definitions:

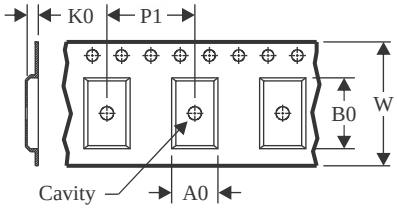
- Enhanced Product - Supports Defense, Aerospace and Medical Applications

## TAPE AND REEL INFORMATION

**REEL DIMENSIONS**

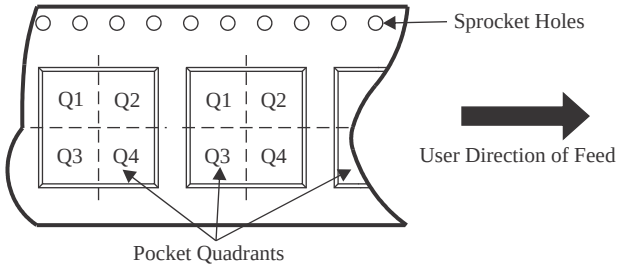



**TAPE DIMENSIONS**




| | |
|---|---|
| A0 | Dimension designed to accommodate the component width |
| B0 | Dimension designed to accommodate the component length |
| K0 | Dimension designed to accommodate the component thickness |
| W | Overall width of the carrier tape |
| P1 | Pitch between successive cavity centers |

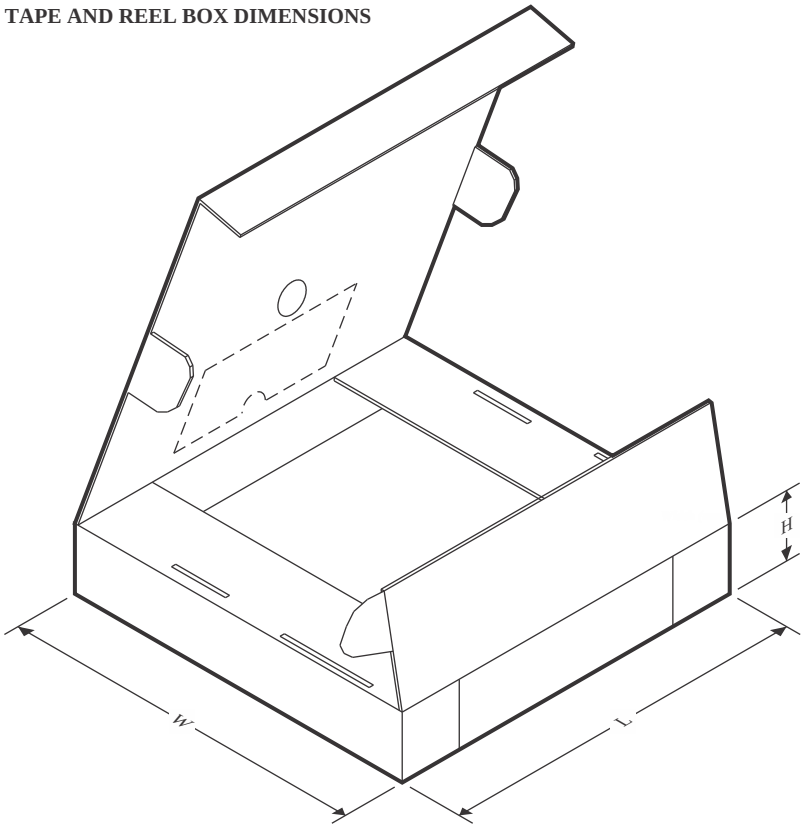
**QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE**




*All dimensions are nominal

| Device | Package Type | Package Drawing | Pins | SPQ | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SN74LVC2G34DBVR | SOT-23 | DBV | 6 | 3000 | 180.0 | 8.4 | 3.2 | 3.2 | 1.4 | 4.0 | 8.0 | Q3 |
| SN74LVC2G34DBVRG4 | SOT-23 | DBV | 6 | 3000 | 178.0 | 9.0 | 3.23 | 3.17 | 1.37 | 4.0 | 8.0 | Q3 |
| SN74LVC2G34DBVT | SOT-23 | DBV | 6 | 250 | 180.0 | 8.4 | 3.23 | 3.17 | 1.37 | 4.0 | 8.0 | Q3 |
| SN74LVC2G34DBVT | SOT-23 | DBV | 6 | 250 | 178.0 | 9.0 | 3.23 | 3.17 | 1.37 | 4.0 | 8.0 | Q3 |
| SN74LVC2G34DBVTG4 | SOT-23 | DBV | 6 | 250 | 178.0 | 9.0 | 3.23 | 3.17 | 1.37 | 4.0 | 8.0 | Q3 |
| SN74LVC2G34DCKR | SC70 | DCK | 6 | 3000 | 178.0 | 9.0 | 2.4 | 2.5 | 1.2 | 4.0 | 8.0 | Q3 |
| SN74LVC2G34DCKR | SC70 | DCK | 6 | 3000 | 180.0 | 8.4 | 2.3 | 2.5 | 1.2 | 4.0 | 8.0 | Q3 |
| SN74LVC2G34DCKRG4 | SC70 | DCK | 6 | 3000 | 178.0 | 9.2 | 2.4 | 2.4 | 1.22 | 4.0 | 8.0 | Q3 |
| SN74LVC2G34DRLR | SOT-5X3 | DRL | 6 | 4000 | 180.0 | 8.4 | 2.0 | 1.8 | 0.75 | 4.0 | 8.0 | Q3 |
| SN74LVC2G34YZPR | DSBGA | YZP | 6 | 3000 | 178.0 | 9.2 | 1.02 | 1.52 | 0.63 | 4.0 | 8.0 | Q1 |

TAPE AND REEL BOX DIMENSIONS




*All dimensions are nominal

| Device | Package Type | Package Drawing | Pins | SPQ | Length (mm) | Width (mm) | Height (mm) |
|---|---|---|---|---|---|---|---|
| SN74LVC2G34DBVR | SOT-23 | DBV | 6 | 3000 | 210.0 | 185.0 | 35.0 |
| SN74LVC2G34DBVRG4 | SOT-23 | DBV | 6 | 3000 | 180.0 | 180.0 | 18.0 |
| SN74LVC2G34DBVT | SOT-23 | DBV | 6 | 250 | 202.0 | 201.0 | 28.0 |
| SN74LVC2G34DBVT | SOT-23 | DBV | 6 | 250 | 180.0 | 180.0 | 18.0 |
| SN74LVC2G34DBVTG4 | SOT-23 | DBV | 6 | 250 | 180.0 | 180.0 | 18.0 |
| SN74LVC2G34DCKR | SC70 | DCK | 6 | 3000 | 180.0 | 180.0 | 18.0 |
| SN74LVC2G34DCKR | SC70 | DCK | 6 | 3000 | 210.0 | 185.0 | 35.0 |
| SN74LVC2G34DCKRG4 | SC70 | DCK | 6 | 3000 | 180.0 | 180.0 | 18.0 |
| SN74LVC2G34DRLR | SOT-5X3 | DRL | 6 | 4000 | 210.0 | 185.0 | 35.0 |
| SN74LVC2G34YZPR | DSBGA | YZP | 6 | 3000 | 220.0 | 220.0 | 35.0 |

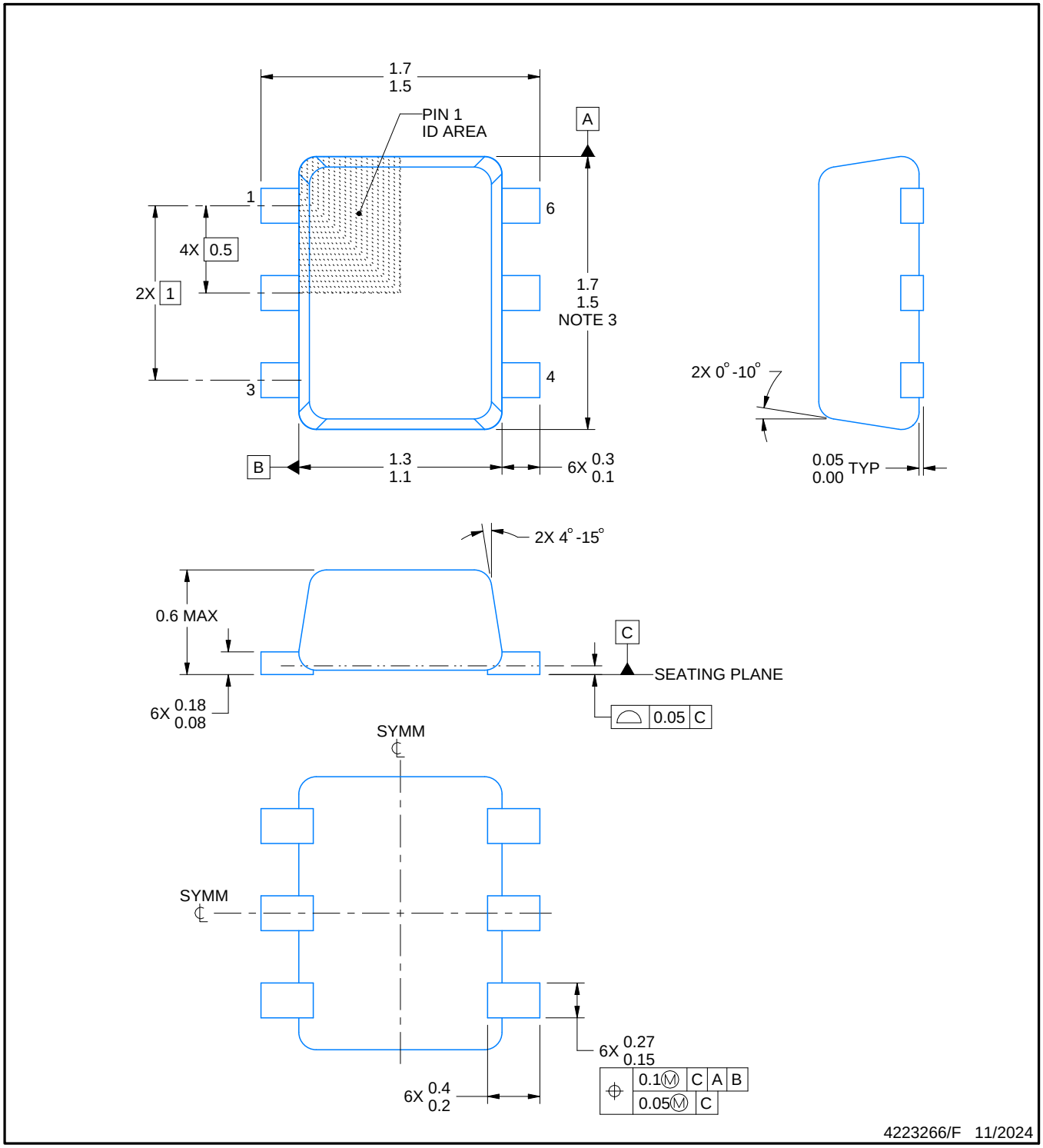


# PACKAGE OUTLINE

## DRL0006A

## SOT - 0.6 mm max height

PLASTIC SMALL OUTLINE




NOTES:

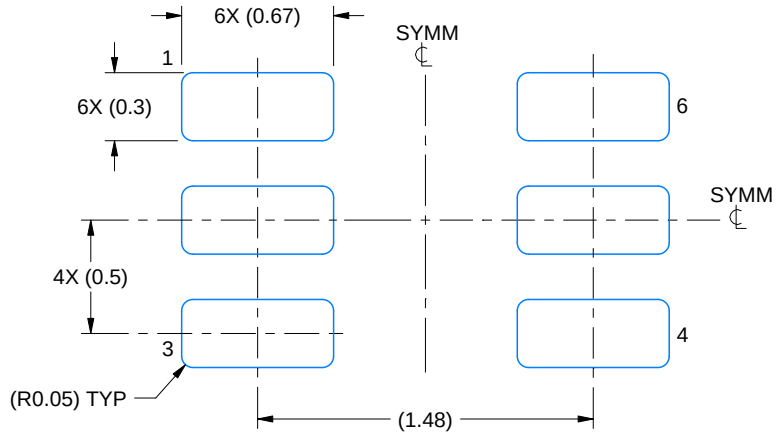
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. Reference JEDEC registration MO-293 Variation UAAD

# EXAMPLE BOARD LAYOUT

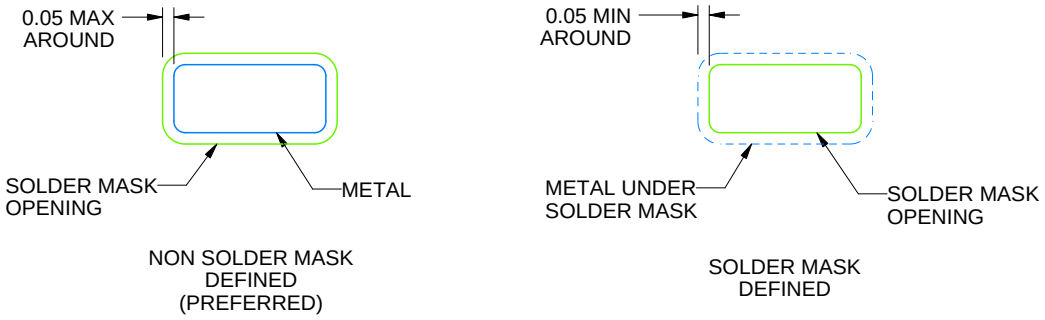
## DRL0006A

## SOT - 0.6 mm max height

PLASTIC SMALL OUTLINE




LAND PATTERN EXAMPLE
SCALE:30X




SOLDERMASK DETAILS

4223266/F 11/2024

NOTES: (continued)

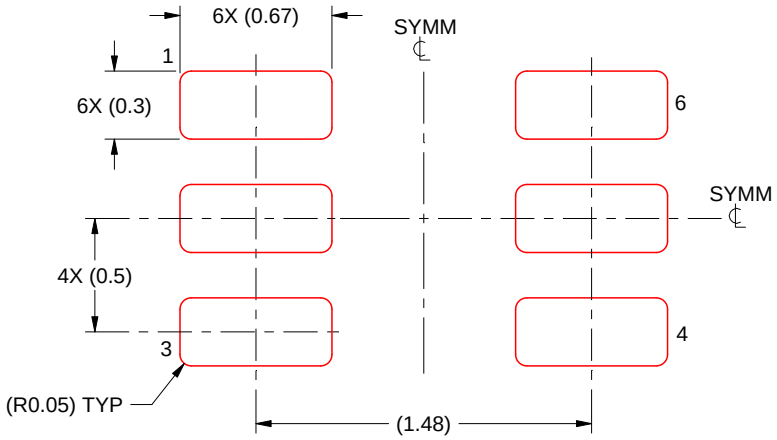
5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.
7. Land pattern design aligns to IPC-610, Bottom Termination Component (BTC) solder joint inspection criteria.

# EXAMPLE STENCIL DESIGN

## DRL0006A

## SOT - 0.6 mm max height

PLASTIC SMALL OUTLINE




SOLDER PASTE EXAMPLE
BASED ON 0.1 mm THICK STENCIL
SCALE:30X

4223266/F 11/2024

NOTES: (continued)

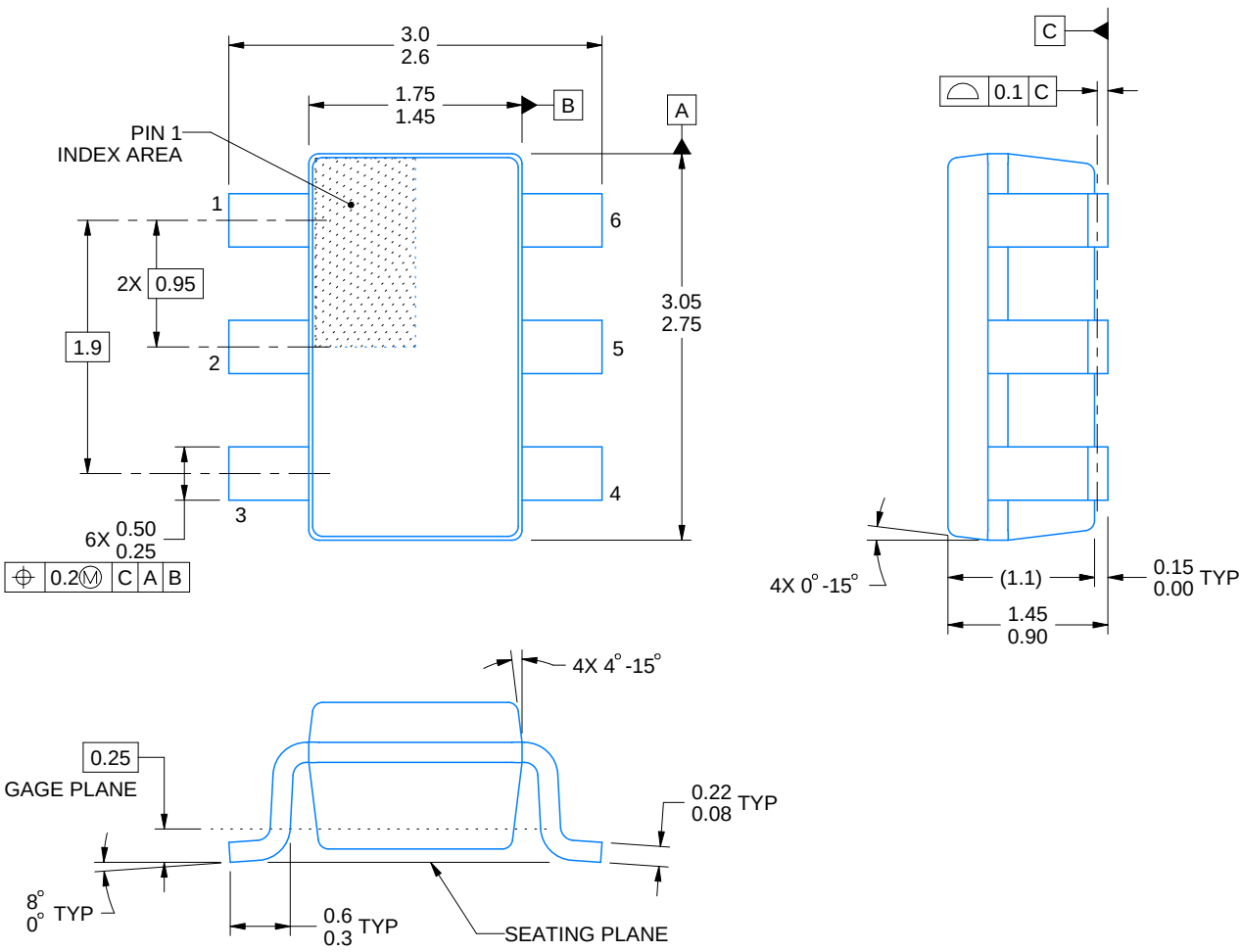
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

# PACKAGE OUTLINE

## DBV0006A

## SOT-23 - 1.45 mm max height

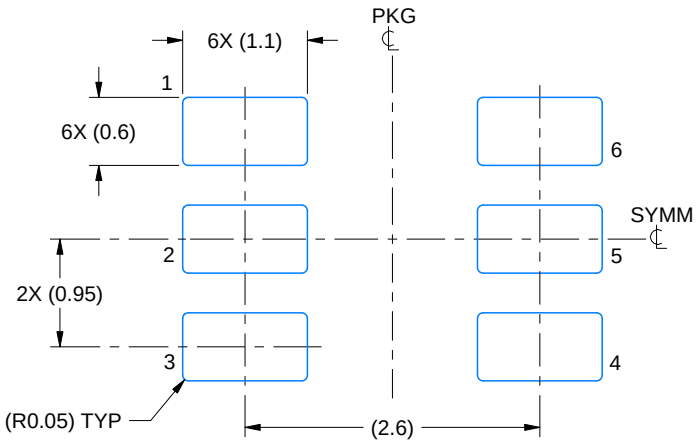
SMALL OUTLINE TRANSISTOR



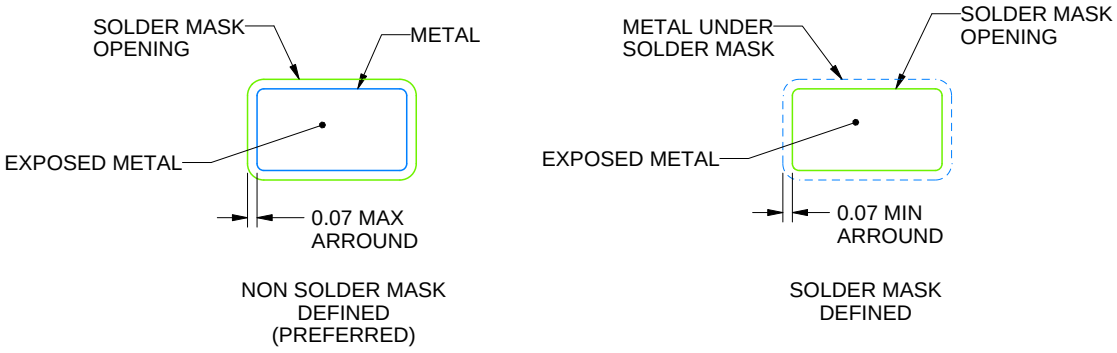

4214840/G 08/2024

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Body dimensions do not include mold flash or protrusion. Mold flash and protrusion shall not exceed 0.25 per side.
4. Leads 1,2,3 may be wider than leads 4,5,6 for package orientation.
5. Refernce JEDEC MO-178.




LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:15X

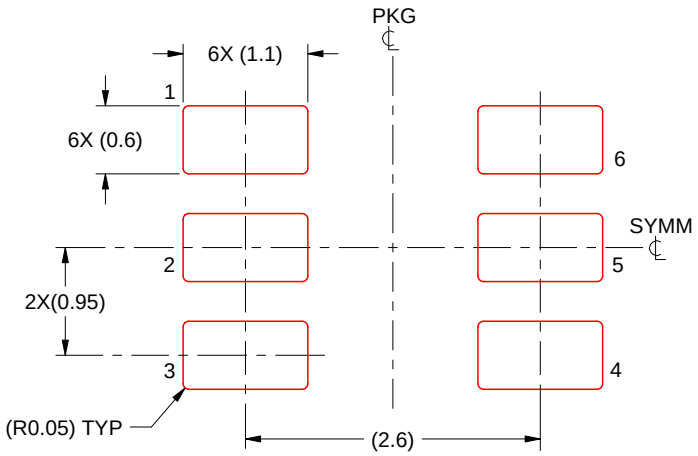



SOLDER MASK DETAILS

4214840/G 08/2024

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.




SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:15X

4214840/G 08/2024

NOTES: (continued)

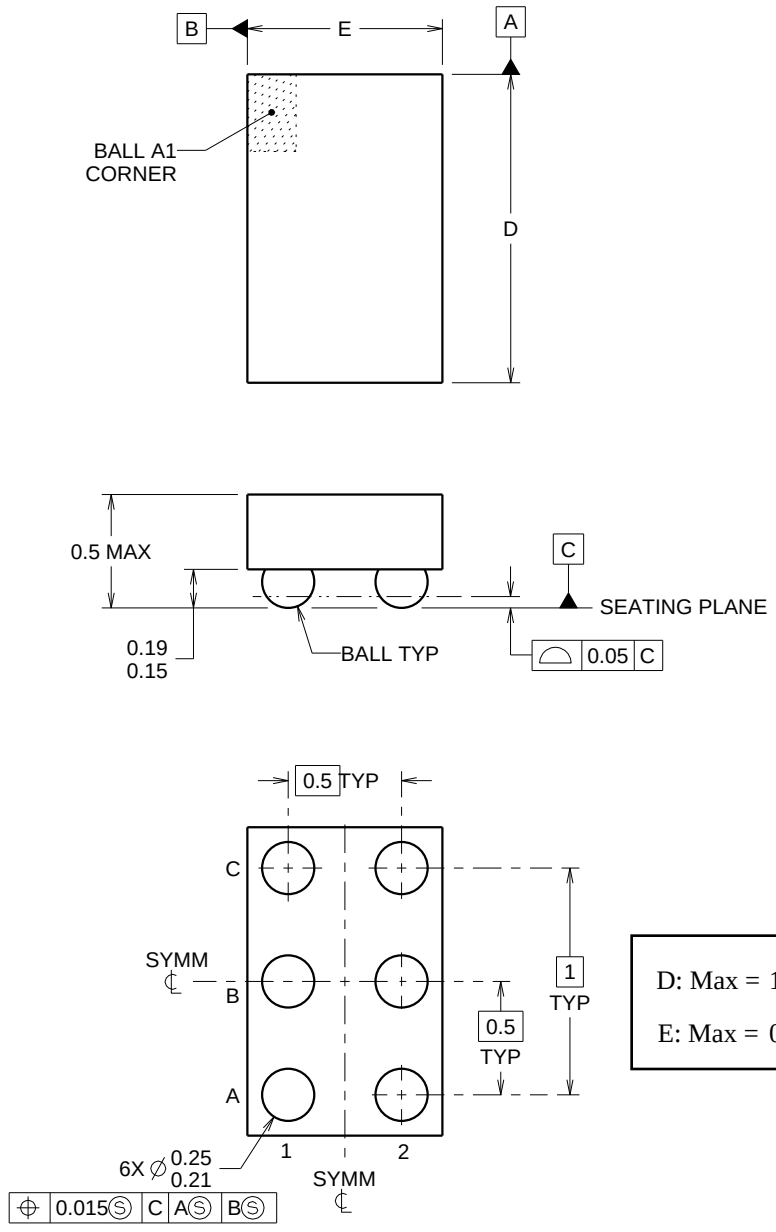
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

# PACKAGE OUTLINE

## YZP0006

## DSBGA - 0.5 mm max height

DIE SIZE BALL GRID ARRAY




D: Max = 1.418 mm, Min =1.358 mm

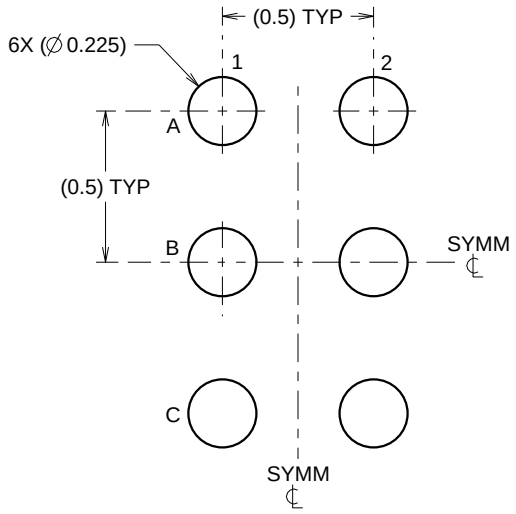
E: Max = 0.918 mm, Min =0.858 mm

4219524/A 06/2014

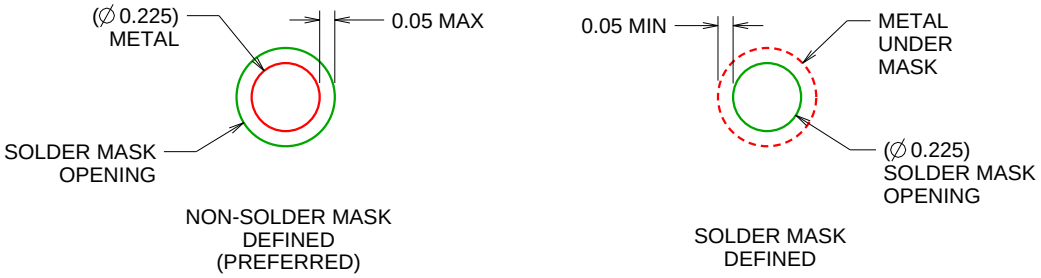
NOTES:

NanoFree Is a trademark of Texas Instruments.

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. NanoFree™ package configuration.




LAND PATTERN EXAMPLE
SCALE:40X




SOLDER MASK DETAILS
NOT TO SCALE

4219524/A 06/2014

NOTES: (continued)

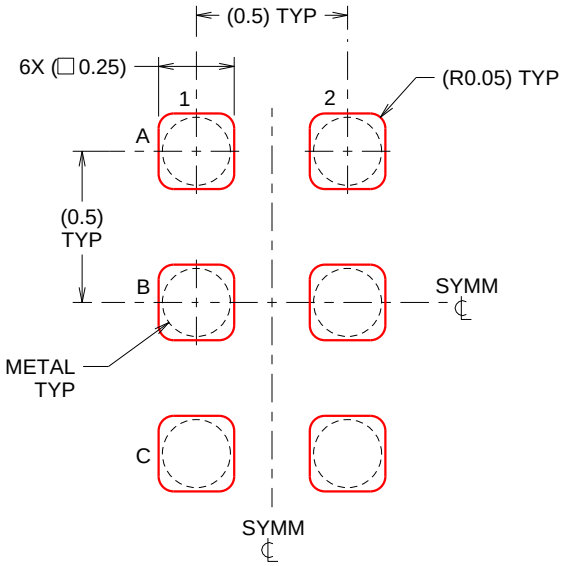
4. Final dimensions may vary due to manufacturing tolerance considerations and also routing constraints. For more information, see Texas Instruments literature number SBVA017 (www.ti.com/lit/sbva017).

# EXAMPLE STENCIL DESIGN

## YZP0006

## DSBGA - 0.5 mm max height

DIE SIZE BALL GRID ARRAY




SOLDER PASTE EXAMPLE
BASED ON 0.1 mm THICK STENCIL
SCALE:40X

4219524/A 06/2014

NOTES: (continued)

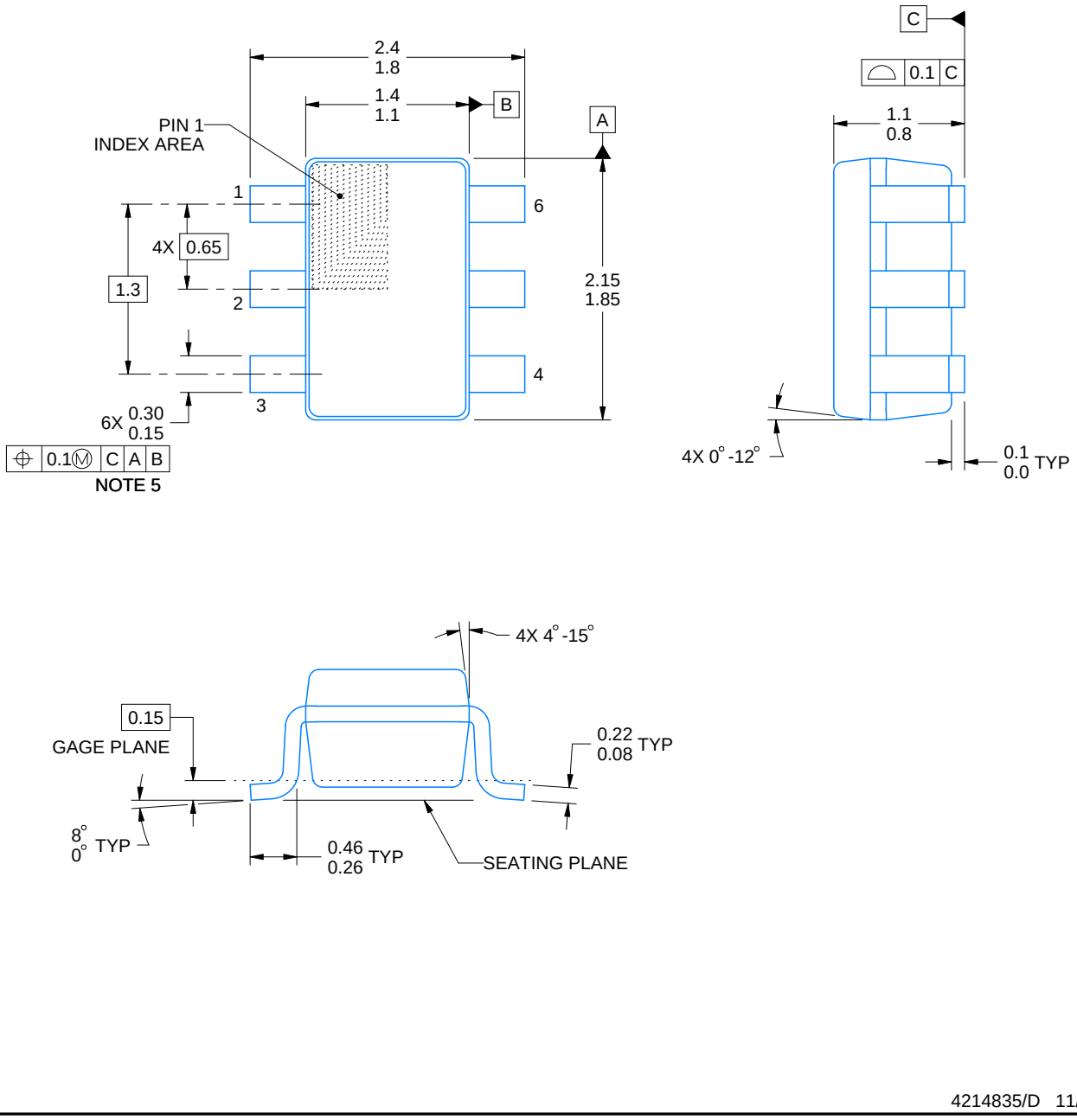
5. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release.

# PACKAGE OUTLINE

## DCK0006A

## SOT - 1.1 max height

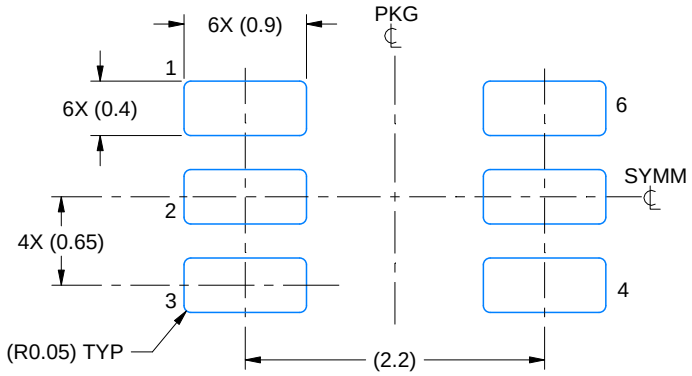
SMALL OUTLINE TRANSISTOR



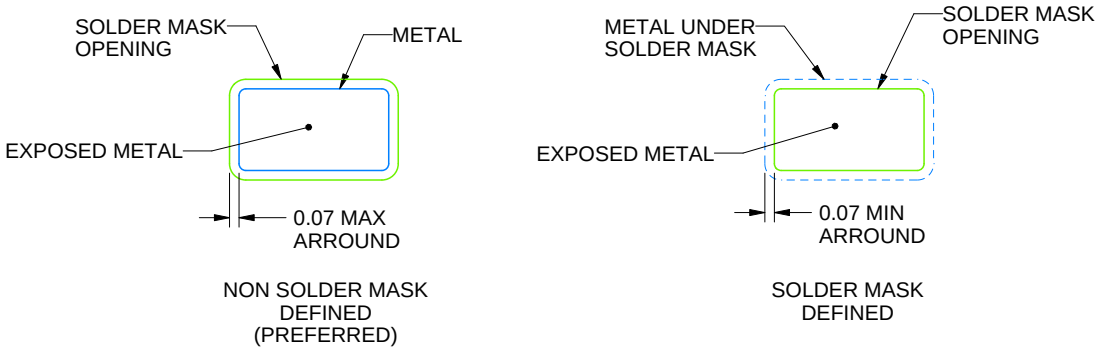

4214835/D 11/2024

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Body dimensions do not include mold flash or protrusion. Mold flash and protrusion shall not exceed 0.15 per side.
4. Falls within JEDEC MO-203 variation AB.




LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:18X

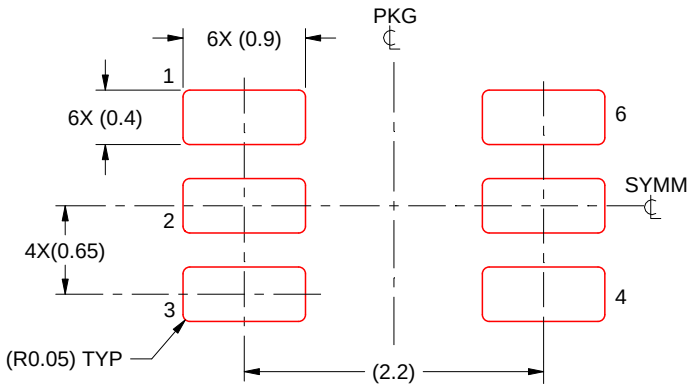



SOLDER MASK DETAILS

4214835/D 11/2024

NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.




SOLDER PASTE EXAMPLE
BASED ON 0.125 THICK STENCIL
SCALE:18X

4214835/D 11/2024

NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

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